

DIY Data Science Curriculum — High-Level Outline

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This document summarizes the high-level course structure we've been building around the restaurant tips dataset. Each session lists practical learning goals and core concepts to revisit.

Session 1 — Environment, Tools, and Tips Dataset Overview

Learning Goals

- Set up Python, Jupyter, conda environment, and project folder structure.
- Load the 'tips' dataset into a pandas DataFrame.
- Understand each column and basic data types.

Key Concepts

- Using conda environments and requirements/environment.yml.
- Reading CSVs with pandas (read_csv).
- Inspecting data: head(), info(), describe().
- Basic data cleaning and renaming columns.

Session 2 — EDA, Feature Engineering, and Simple Visuals

Learning Goals

- Create new features like tip percentage of bill.
- Explore distributions and relationships with simple plots.
- Get comfortable moving between code, charts, and interpretation.

Key Concepts

- Feature engineering (tip_pct = tip_usd / bill_total_usd).
- Histograms, bar plots, and boxplots (seaborn/matplotlib).
- Grouping and aggregation with groupby().
- Writing ELI5-style interpretations of plots.

Session 3 — Correlation and Simple Linear Regression

Learning Goals

- Quantify linear relationships between numeric variables.
- Fit a simple linear regression: tip_usd ~ bill_total_usd.
- Interpret slope, intercept, and R-squared in plain language.

Key Concepts

- Pearson correlation vs. Spearman correlation.
- Scatter plots with trend lines (sns.lmplot / regplot).
- scikit-learn LinearRegression: fit, predict, coef_, intercept_.
- Meaning of R^2 and adjusted R^2 .

Session 4 — Multiple Regression and Categorical Variables

Learning Goals

- Add additional predictors like `party_size`, `gender`, `is_smoker`.
- Encode categorical variables correctly for regression.
- Compare simple vs. multiple regression models.

Key Concepts

- Dummy variables (one-hot encoding) for gender and smoker.
- Reading regression coefficients while holding others constant.
- Standardized coefficients for feature importance.
- Train/test split for basic model validation.

Session 5 — Residual Diagnostics and Assumptions

Learning Goals

- Check if linear regression assumptions are roughly satisfied.
- Use residual plots to look for patterns and problems.
- Understand normality, homoscedasticity, and endogeneity at a high level.

Key Concepts

- Residuals vs predicted plots (looking for random cloud around 0).
- Residual histograms and Q–Q plots for normality.
- Homoscedasticity vs heteroscedasticity.
- Endogeneity (feature correlated with error term) and why it biases OLS.

Session 6 — Transformations and Robustness

Learning Goals

- Experiment with log-transforming the target ($\log(\text{tip})$).
- Compare linear vs log-transformed models using RMSE and R^2 .
- See when a transformation helps or does not help.

Key Concepts

- Log and $\log_{1p}$ transforms for skewed variables.
- Interpreting coefficients in log models in plain language.
- RMSE as a scale-dependent error metric.
- Trade-offs: better residual shape vs easier interpretability.

Session 7 — Multicollinearity and VIF

Learning Goals

- Detect when predictors are telling the same story.
- Compute Variance Inflation Factor (VIF) for each feature.
- Decide when multicollinearity is a concern and what to do about it.

Key Concepts

- Correlation matrix heatmaps for features.
- Variance Inflation Factor (VIF) and typical thresholds.
- Impact of multicollinearity on coefficient stability.
- Strategies: dropping features, combining features, or using regularization (preview).

Session 8 — Model Comparison and Baselines

Learning Goals

- Define a simple baseline model to compare against.
- Compare multiple models using the same metrics.
- Summarize model performance in a small, clear table and narrative.

Key Concepts

- Baselines (e.g., always predict mean tip).
- Train/test splits and why train performance alone is not enough.
- Comparing RMSE, MAE, and R^2 across models.
- Writing short, stakeholder-friendly conclusions.

Session 9 — Communication and Storytelling with Data

Learning Goals

- Turn models and plots into a coherent story.
- Choose appropriate charts for the question you're answering.
- Practice writing ELI5 explanations for non-technical readers.

Key Concepts

- Framing analysis around questions, not just methods.
- Chart selection: when to use scatter, boxplot, bar, line, etc.
- Annotating charts and adding takeaways in titles/captions.
- Structuring a short written summary (context → findings → implications).

Session 10 — Capstone: Mini Case Study on Tipping Behavior

Learning Goals

- Pull together EDA, modeling, and diagnostics into one workflow.
- Produce a short written report or notebook telling the tipping story.
- Identify next questions and limitations of the current analysis.

Key Concepts

- End-to-end workflow: load → clean → explore → model → diagnose → communicate.
- Version control with Git/GitHub for notebooks and reports.
- Saving figures and using a simple plot index for navigation.
- Reflecting on what you've learned and where to go next.